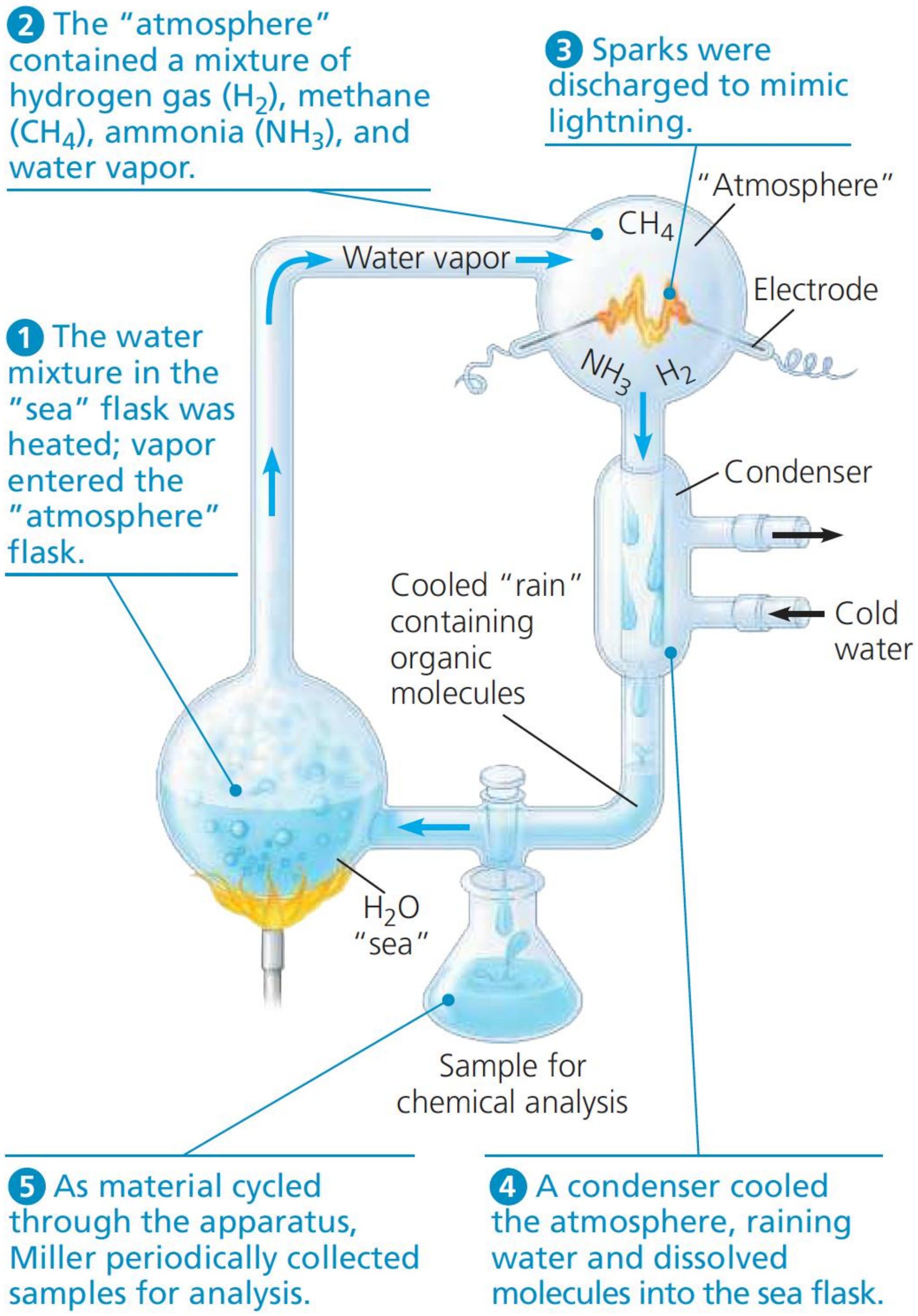
Evolution & Diversity

Origin of Life

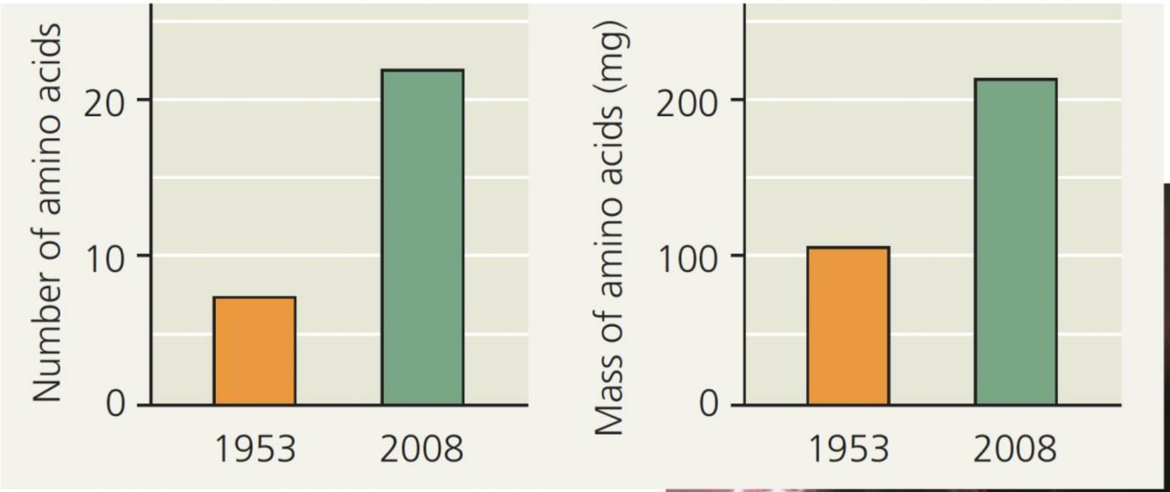
1. The abiotic (nonliving) synthesis of small organic molecules, such as amino acids and nitrogenous bases



Origin of Life

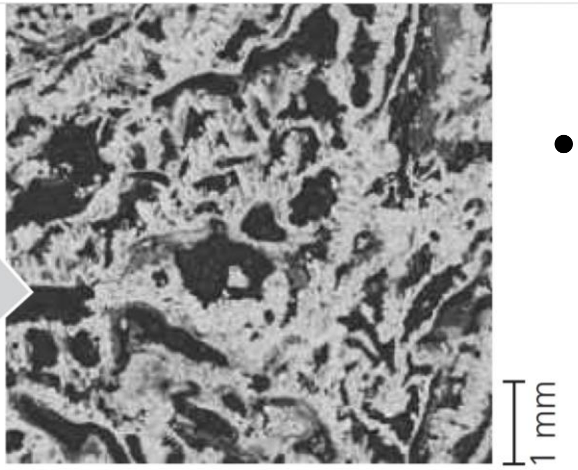
- However, the primitive atmosphere was made up primarily of nitrogen and carbon dioxide and was neither reducing nor oxidizing
- **Volcanic eruptions** initiated life

▼ **Figure 25.2 Amino acid synthesis in a simulated volcanic eruption.** In addition to his classic 1953 study, Miller also conducted an experiment simulating a volcanic eruption. In a 2008 reanalysis of those results, researchers found that far more amino acids were produced under simulated volcanic conditions than were produced in the conditions of the classic 1953 study.



MAKE CONNECTIONS *Explain how more than 20 amino acids could have been obtained in the 2008 study. (See Concept 5.4.)*

- **Alkaline vents** in deep-sea initiated life

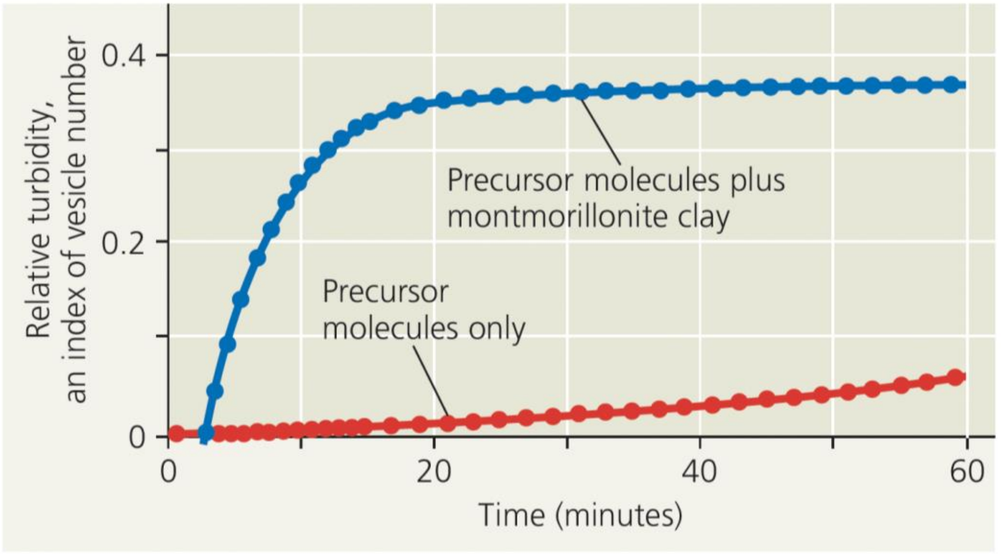


◀ **Figure 25.3 Did life originate in deep-sea alkaline vents?** The first organic compounds may have arisen in warm alkaline vents similar to this one from the 40,000-year-old "Lost City" vent field in the mid-Atlantic Ocean. These vents contain hydrocarbons and are full of tiny pores (inset) lined with iron and other catalytic minerals. Early oceans were acidic, so a pH gradient would have formed between the interior of the vents and the surrounding ocean water. Energy for the synthesis of organic compounds could have been harnessed from this pH gradient.

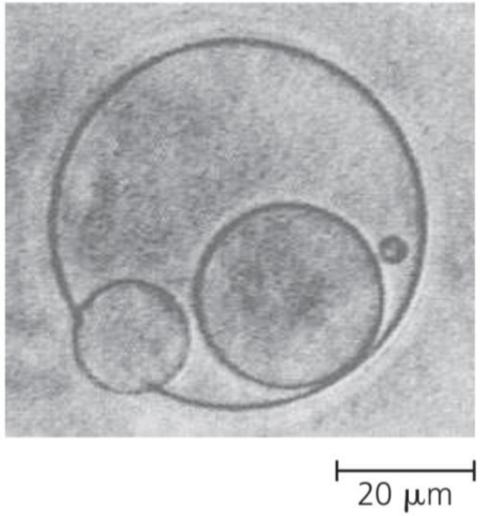
- Life came on Earth came **from outer space**
- 4.5 billion-year-old Murchison meteorite in Australia
- More than 80 types of Amino Acids

Protocell

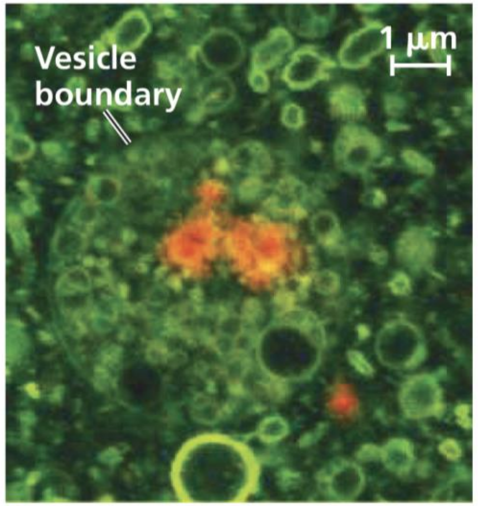
- All organisms must be able to carry out both reproduction and energy processing (metabolism)
- The necessary conditions may have been met in *vesicles*, fluid-filled compartments enclosed by a membrane-like structure (Protocell)
- Vesicles can form spontaneously when lipids are added to water



(a) Self-assembly. The presence of montmorillonite clay greatly increases the rate of vesicle self-assembly.



(b) Reproduction. Vesicles can divide on their own, as in this vesicle "giving birth" to smaller vesicles (LM).

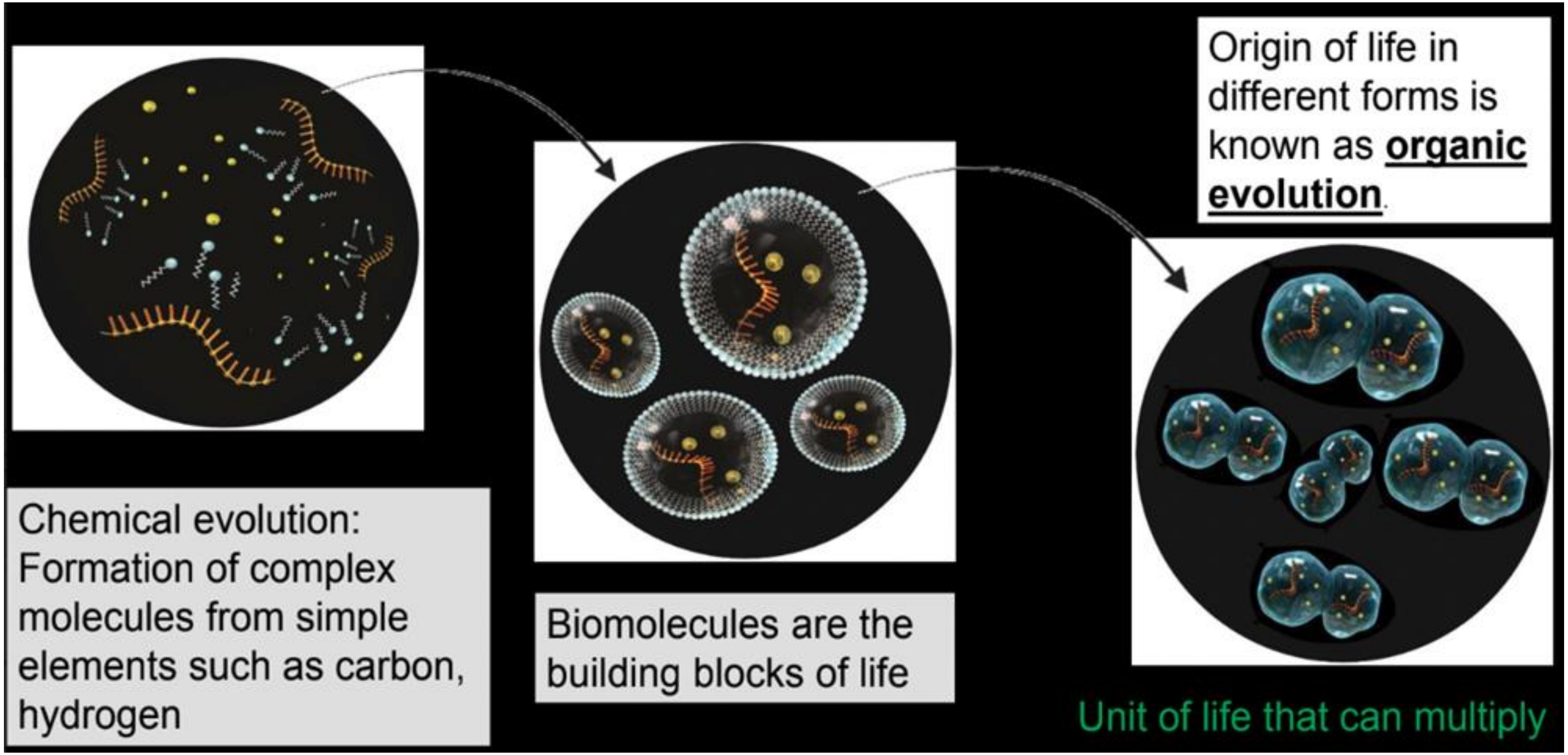


(c) Absorption of RNA. This vesicle has incorporated montmorillonite clay particles coated with RNA (orange).

Protocell

- The first genetic material was most likely RNA, not DNA
- Some RNAs can make complementary copies of a short piece of RNA, provided that they are supplied with nucleotide
- They are called **Ribozymes**
- RNA could have provided the template on which DNA nucleotides were assembled
- DNA is more stable and replicates with less error
- The appearance of DNA set the stage for the blossom of different forms of life that we see in fossils

From biomolecules to Origin of Life



Origin of life: Evolution by selection

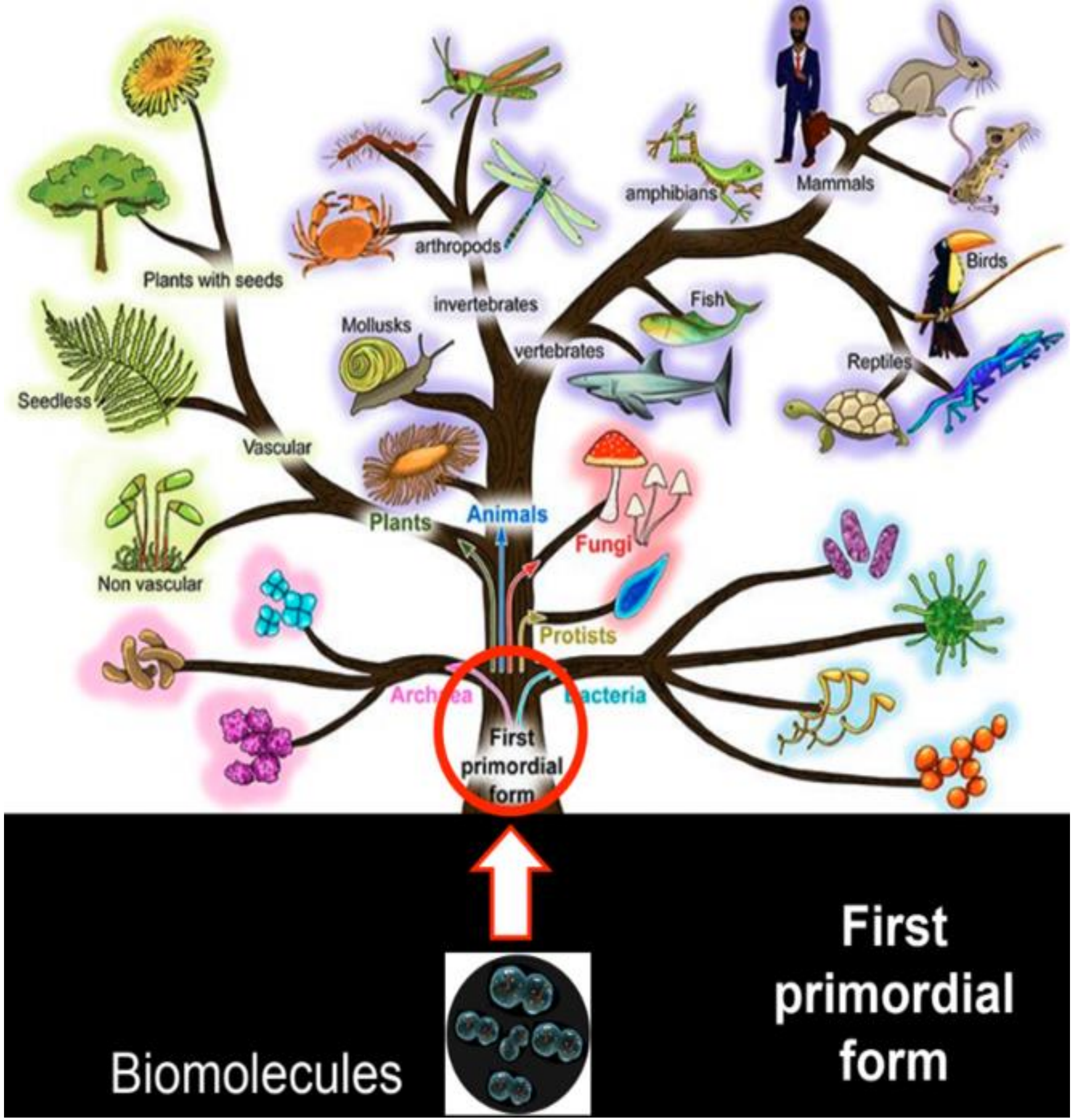
Evolution results in vast diversity in life forms

- All organisms evolved from a single common ancestor

- By slowly accumulating small changes over ~3.8 billion years - Evolution

- ~8 million species evolved on earth

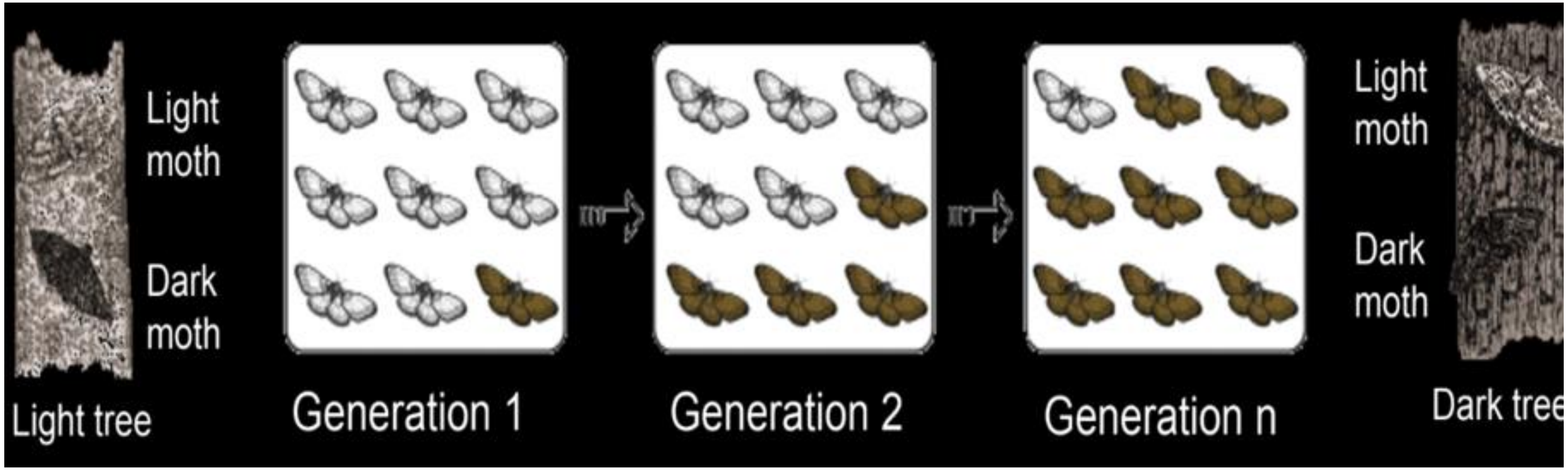
- By studying evolution:
    - Origin of life forms on earth
    - How different life forms have evolved in their specific environment?



Why there is so much diversity in the living organisms?

Evolution:

Gradual change in population characteristics over a period of time



Changes over a small timescale (Micro-evolution)

Over longer time, emergence of new species from common ancestor (Macro-evolution)

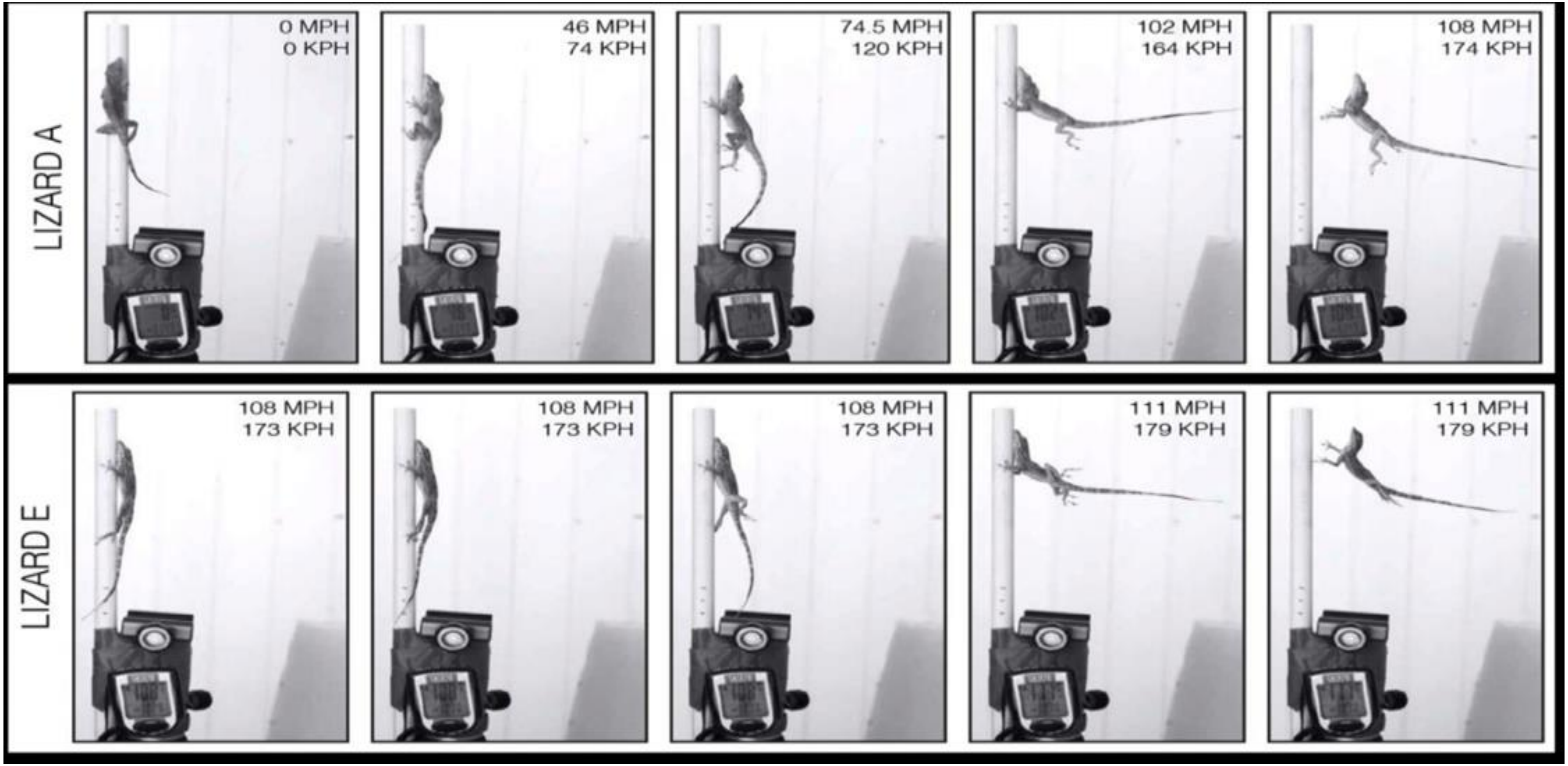
How to understand evolution?

Scientists set up a wind machine to test how well anoles could hold on during a hurricane

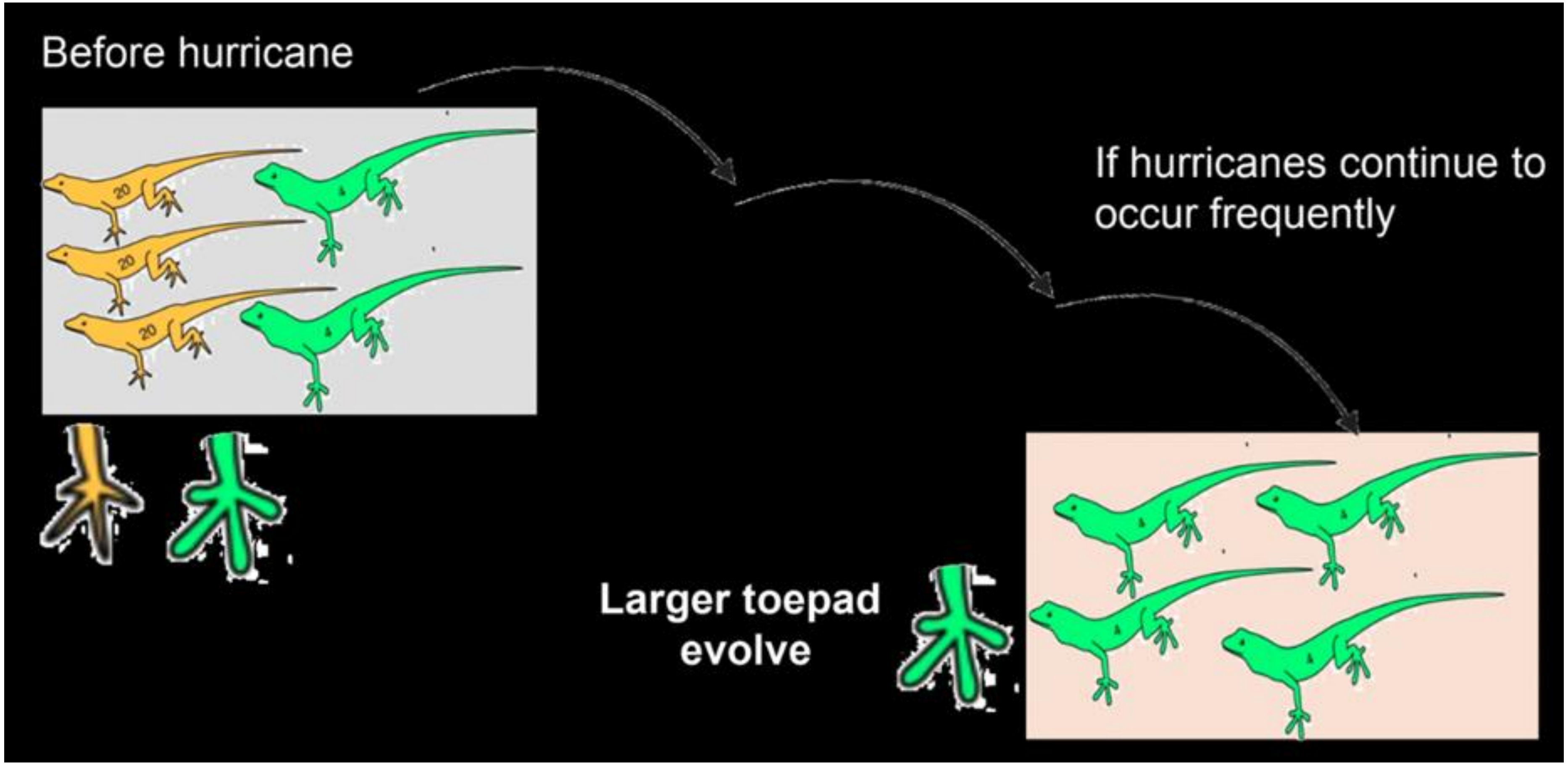
Identify the features that are helping the lizard hold on

https://www.youtube.com/watch?v=aKvKd1SozOc

How to understand evolution?
Observe the pattern



How to connect this to evolution?



How to study evolution?

We can study evolution by measuring gradual change in organisms happening over a course of time

Key definitions in the process of evolution

The scientists measured different physical characters of the lizards such as Toepad size , Forelimb, Hindlimb length – **Traits (Evidence)**

- Different lizards had different Toepad size , Forelimb, Hindlimb length (**Variation**)

- Lizards with different characters responded differently to hurricane

- Hurricane: Force causing change in the lizard characters – lizards with big toepads survived

- Process by which organisms with favorable characters are chosen – **Selection**

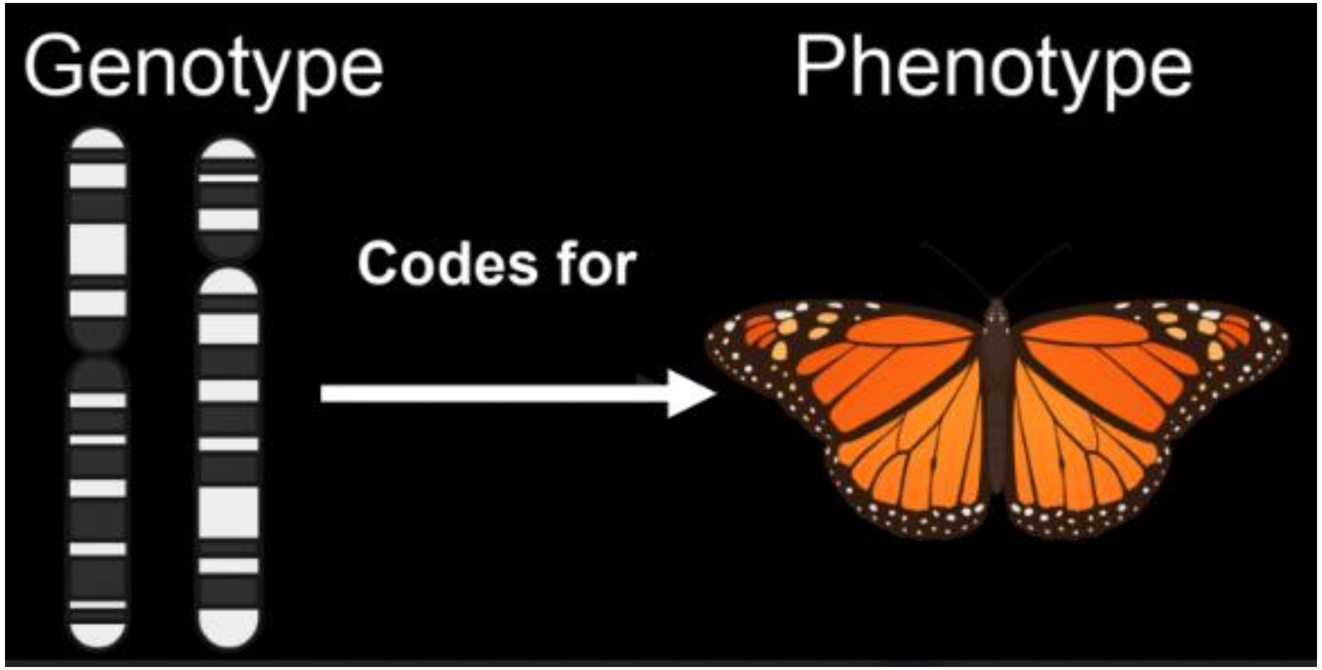
- Many years of such change results in evolution of specific characters

Let us understand it one at a time!

Traits: Physical attributes of an organism, eg. Size, hair color, leaf shape

- **Phenotype**: Collection of characteristics or physical appearance of an organism, e.g. blood type, brown eyes, black fur, pink flowers

- **Genotype**: Collection of all genes



Framework for evolution:

Selection as an evolutionary force

Fundamental principles of selection

1. Organisms within population exhibit variation in traits

2. Some traits are passed on from parents to offspring i.e. variation is heritable

3. Differential survival and reproduction within a population (**fitness**)

4. Variation influences fitness. More ‘fit’ phenotypes will be favored by selection

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This is an example of.....???

All individuals are not alike (phenotypic variation)

Why is variation so important?

If cauliflower goes out of season and there is none available then what will happen to these restaurants?

Variation increases/expands the 'menu' from which selection can choose

Safety net which ensures that certain individuals can adapt to sudden changes

For example, variation in toepad size helped certain lizards survive better during hurricane.

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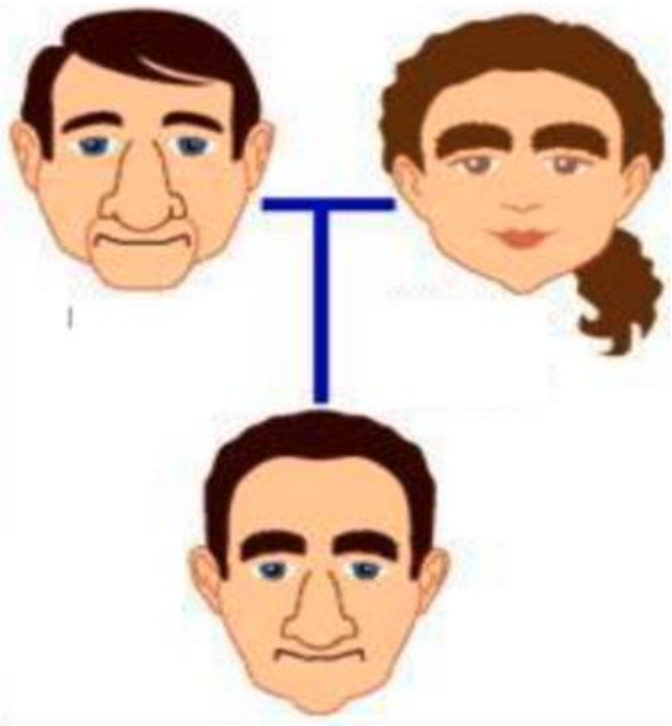
Traits are inherited from parents



Physical traits are coded in .... that offspring receives from parents

- Offspring inherit majority of physical traits from their parents
- **Heritability**

Heritability

Toepad size and length of limbs: heritable (passed to next generation)

- Advantage in hurricane winds

Important: For selection to work, a trait must be passed from parents to offspring i.e. trait must be heritable

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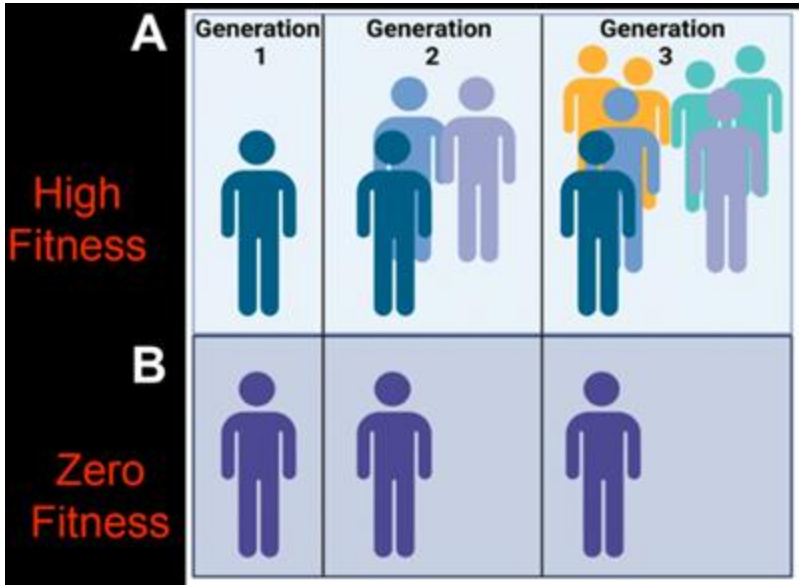
Fitness

Fitness = Lifespan of an individual (Survival)
+ Offspring produced (Reproduction)



- Person A produce more children than B
- But person B goes to gym, has no diseases

Fitness is defined by how long an organism live and how many offspring/progeny an organism produces

Individuals vary in survival and reproduction:
Variation in heritable traits which <u>affect survival and reproduction</u> are most important from selection's 'point of view'

Fundamental principles of selection

1. Organisms within population exhibit <u>variation in traits</u>

2. Some traits are passed on from parents to offspring i.e. <u>variation is heritable</u>

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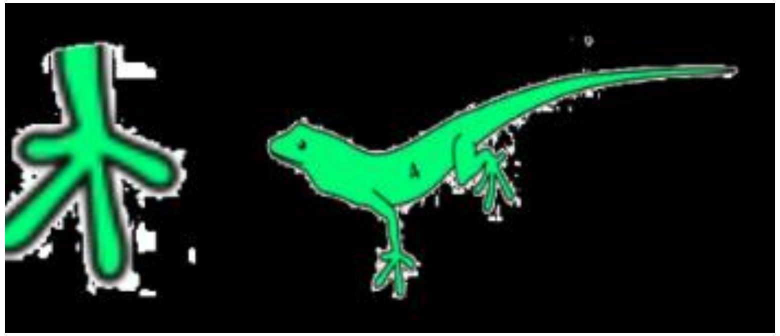
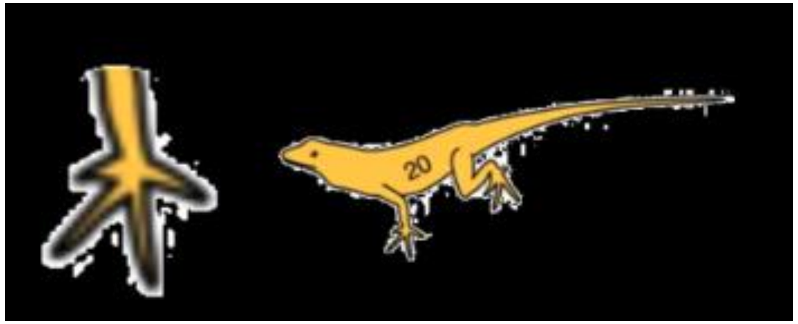
Variation in traits affect survival and reproduction

Some characteristics have advantages
- Large toepad
- – selective advantage
- – better survival
- – more opportunity to produce offspring
- Being large toepad (trait) means higher fitness



Greater reproductive success → more children/offspring



Small toepad
- – higher survival risk
- – lower fitness

Lower reproductive success → fewer offspring/children

Several characters can determine fitness

What is common among these?

Common ancestor
&
Artificial evolution

What is Artificial selection?

Modification of a species through human actions

“evolutionary changes occurring as a consequence of conditions imposed by the experimenter” (e.g by human in the Lab ~ experimental selection)

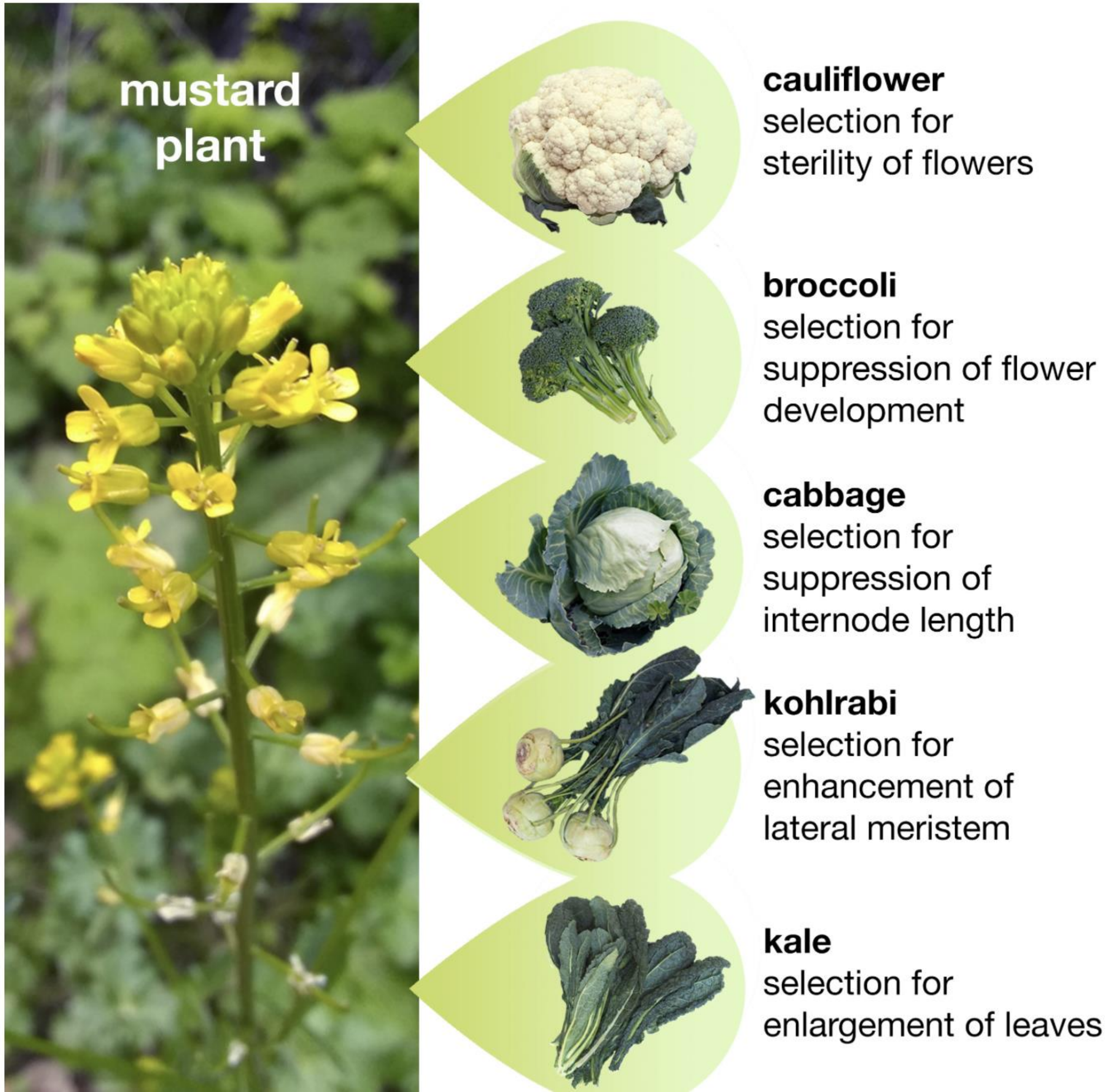
Conditions: Temperature, Food source, Infection

Selective breeding of organisms to produce offspring with specific traits

Artificial selection

Modification of a species <u>through human actions</u>

Selective breeding of organisms to produce offspring with specific traits



Generation Time frame

By the time you graduate (4 years)

Bacteria (E.coli) – 20 minutes (70,000 generations)

Yeast – 80 minutes (26,000 generations)

Worm (C. elegans) – 3 days (486 generations)

Fly (Drosophila) – 12 days (120 generations)

Monkey – 4 years (1 generation)

Why to study evolution?

Evolution can be used as an engineering tool

Evolution continually create …. Take examples to create

Allows you to discover relationship between a trait and environment

Can we use evolution to solve a commercial problem?

Global baking industry ~ 100 million ton (2023)

Requirement hard to meet
Baker's yeast has low stress tolerance
Difficult to use yeast efficiently
Unable to meet the demand

Artificial selection can be used to make high stress tolerant yeast in high yield

Bibliography

Biology- A global Approach: Campbell [Chapters 25 & 26]

Natural selection in a hurricane: The lizards that won't let go

https://www.youtube.com/watch?v=aKvKd1SozOc

The Phylogenetic Tree of Anole Lizards — HHMI BioInteractive Video

https://www.youtube.com/watch?v=rdZOwyDbyL0